Probability and a fraud-detection simulation

Homework 3

Xinchen Leng

Part 1 works the four analytical problems in full. Part 2 is the simulation of Question 1. The code, with seed 42 and 100,000 transactions, is at the GitHub link below. That link is the Part 2 deliverable.

https://github.com/Orlandoxinchen/homework-3

Part 1. Analytical problems

1. Amazon fraud detection

The system has sensitivity P(Flagged | Fraud) = 0.98, specificity P(Not flagged | Legitimate) = 0.95, and base rate P(Fraud) = 0.001. Specificity fixes the false-positive rate:

$$P(\text{Flagged} \mid \text{Legitimate}) = 1 - 0.95 = 0.05.$$

$$P(\text{Legitimate}) = 1 - 0.001 = 0.999.$$

1(a). Probability that a flagged transaction is fraud

Bayes' theorem writes the target as the joint probability of a true detection divided by the probability of any flag:

$$P(\text{Fraud} \mid \text{Flagged}) = P(\text{Flagged} \mid \text{Fraud})\, P(\text{Fraud}) / P(\text{Flagged}).$$

The law of total probability splits flags into those that come from fraud and those that come from legitimate transactions:

$$P(\text{Flagged}) = P(\text{Flagged} \mid \text{Fraud})\, P(\text{Fraud}) + P(\text{Flagged} \mid \text{Legitimate})\, P(\text{Legitimate})$$

$$= (0.98)(0.001) + (0.05)(0.999)$$

$$= 0.00098 + 0.04995$$

$$= 0.05093.$$

The numerator is the first of those two pieces, the true detections. Dividing gives

$$P(\text{Fraud} \mid \text{Flagged}) = 0.00098 / 0.05093 = 98/5093 \approx 0.01924.$$

Answer. 0.01924, or about 1.92 percent. A flag is weak evidence of fraud. Fraud is rare, and a 5 percent false-positive rate applied to the 99.9 percent of transactions that are legitimate produces almost all of the flags.

1(b). Specificity raised to 0.99

Specificity 0.99 means P(Flagged | Legitimate) = 0.01. Sensitivity and the base rate stay the same, so the true-detection term is unchanged:

$$P(\text{Flagged}) = (0.98)(0.001) + (0.01)(0.999)$$

$$= 0.00098 + 0.00999 = 0.01097.$$

$P(Fraud \mid Flagged) = 0.00098 / 0.01097 = 98/1097 \approx 0.08933.$

Answer. 0.08933, or about 8.93 percent. The posterior is about 4.6 times the answer in part (a).

The business change is in the review queue, not in the share of fraud that is caught. Sensitivity is still 0.98, so the same fraction of fraudulent transactions is flagged. What falls is the false-positive rate, from 5 percent to 1 percent. Flags drop from about 5.09 percent of all transactions to about 1.10 percent, and a reviewed flag is much more likely to be a real case. Automatic blocking is still a poor rule: even after the change, about 91 percent of flags are legitimate. The gain is a shorter queue and fewer good customers pulled into review.

2. Portfolio of two independent tech stocks

Returns are written as decimals. Stock A and Stock B are independent, so Cov(X, Y) = 0.

Stock A return	−2%	0%	3%	5%
P(X)	0.1	0.3	0.4	0.2
Stock B return	3%	1%	4%	
P(Y)	0.2	0.5	0.3	

2(a). Expected returns

$E[X] = (-0.02)(0.1) + (0)(0.3) + (0.03)(0.4) + (0.05)(0.2)$

$= -0.002 + 0 + 0.012 + 0.010 = 0.020.$

$E[Y] = (0.03)(0.2) + (0.01)(0.5) + (0.04)(0.3)$

$= 0.006 + 0.005 + 0.012 = 0.023.$

Answer. E[X] = 0.020 (2.00 percent). E[Y] = 0.023 (2.30 percent).

2(b). Variances, shortcut formula

The shortcut is $Var(X) = E[X^2] - (E[X])^2$.

$E[X^2] = (-0.02)^2(0.1) + (0.03)^2(0.4) + (0.05)^2(0.2)$

$= (0.0004)(0.1) + (0.0009)(0.4) + (0.0025)(0.2)$

$= 0.00004 + 0.00036 + 0.00050 = 0.00090.$

$Var(X) = 0.00090 - (0.020)^2 = 0.00090 - 0.00040 = 0.00050.$

$E[Y^2] = (0.03)^2(0.2) + (0.01)^2(0.5) + (0.04)^2(0.3)$

$= (0.0009)(0.2) + (0.0001)(0.5) + (0.0016)(0.3)$

$= 0.00018 + 0.00005 + 0.00048 = 0.00071.$

$Var(Y) = 0.00071 - (0.023)^2 = 0.00071 - 0.000529 = 0.000181.$

Answer. Var(X) = 0.00050. Var(Y) = 0.000181.

2(c). Portfolio R = 0.6X + 0.4Y

Linearity of expectation does not need independence. Independence is used for the variance, where the covariance term drops out.

$E[R] = 0.6\,E[X] + 0.4\,E[Y] = 0.6(0.020) + 0.4(0.023)$

$= 0.012 + 0.0092 = 0.0212.$

$Var(R) = (0.6)^2\,Var(X) + (0.4)^2\,Var(Y)$

$= (0.36)(0.00050) + (0.16)(0.000181)$

$= 0.000180 + 0.00002896 = 0.00020896.$

Answer. E[R] = 0.0212 (2.12 percent). Var(R) = 0.00020896.

3. Server response time

The density is $f(x) = (3/8)\,x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise.

3(a). Valid PDF

On [0, 2], $x^2 \ge 0$ and $3/8 > 0$, so $f(x) \ge 0$. Outside that interval the density is zero, which is also nonnegative. The integral is

$\int (3/8)\,x^2\,dx \text{ from } 0 \text{ to } 2 = (3/8) \cdot (x^3/3) \text{ from } 0 \text{ to } 2$

$= (1/8)\,x^3 \text{ from } 0 \text{ to } 2 = (1/8)(8) - 0 = 1.$

Answer. The function is nonnegative and integrates to 1, so it is a valid PDF.

3(b). CDF on [0, 2]

$F(x) = P(X \le x) = \int (3/8)\,t^2\,dt \text{ from } 0 \text{ to } x$

$= (1/8)\,t^3 \text{ from } 0 \text{ to } x = x^3/8, \quad 0 \le x \le 2.$

Answer. $F(x) = x^3/8$ for $0 \le x \le 2$.

3(c). Three probabilities

$P(X \le 1) = F(1) = 1^3/8 = 1/8 = 0.125.$

$P(0.5 \le X \le 1.5) = F(1.5) - F(0.5) = (1.5)^3/8 - (0.5)^3/8$

$= (3.375)/8 - (0.125)/8 = 3.250/8 = 0.40625.$

In fractions, $(1.5)^3 = 27/8$ and $(0.5)^3 = 1/8$, so

$F(1.5) - F(0.5) = (27/64) - (1/64) = 26/64 = 13/32.$

$P(X > 1.5) = 1 - F(1.5) = 1 - 27/64 = 37/64 = 0.578125.$

Answer. $P(X \le 1) = 1/8 = 0.125$. $P(0.5 \le X \le 1.5) = 13/32 = 0.40625$. $P(X > 1.5) = 37/64 = 0.578125$.

4. Binomial and hypergeometric

4(a). Binomial clicks

Each of 10 users clicks with probability 0.20, independently, so X ~ Binomial(n = 10, p = 0.20). The point mass is

$$P(X = 3) = C(10, 3)\ (0.20)^3\ (0.80)^7.$$

$$C(10, 3) = 120.$$

$$(0.20)^3 = 0.008.$$

$$(0.80)^7 = 0.2097152.$$

$$P(X = 3) = 120 \times 0.008 \times 0.2097152 = 0.201326592.$$

The same value as a reduced fraction is 393,216 / 1,953,125, from $C(10, 3)\ (1/5)^3\ (4/5)^7$.

Answer. P(X = 3) = 0.201326592.

4(b). Hypergeometric defects

The batch has N = 20 products, of which K = 4 are defective. The inspector draws n = 5 products without replacement, so Y ~ Hypergeometric(N = 20, K = 4, n = 5). Exactly one defective item in the sample means 1 defective chosen from 4 and 4 non-defective chosen from 16:

$$P(Y = 1) = C(4, 1)\ C(16, 4) / C(20, 5).$$

$$C(4, 1) = 4.$$

$$C(16, 4) = (16 \times 15 \times 14 \times 13) / (4 \times 3 \times 2 \times 1) = 1{,}820.$$

$$C(20, 5) = (20 \times 19 \times 18 \times 17 \times 16) / 120 = 15{,}504.$$

$$P(Y = 1) = (4 \times 1{,}820) / 15{,}504 = 7{,}280 / 15{,}504 = 455/969 \approx 0.46956.$$

Answer. P(Y = 1) = 455/969 ≈ 0.46956.

Part 2. Coding problem

The simulation follows the assignment steps: seed 42, 100,000 transactions, fraud drawn at rate 0.001, fraudulent transactions flagged with probability 0.98, and legitimate transactions flagged with probability 0.05. The file is *fraud_simulation.py*.

https://github.com/Orlandoxinchen/homework-3

With this seed the draw contains 102 fraudulent transactions, not the expected 100. Of those 102, 98 are flagged. Legitimate transactions contribute 4,894 flags. The empirical conditional probability is therefore

$$P(Fraud \mid Flagged) = 98 / 4{,}992 = 0.0196,$$

printed to four decimals, against the Part 1(a) value 0.0192. The two agree to three decimals. The gap is ordinary binomial noise in a rare-event sample: only about 100 frauds are expected, so the numerator moves by a few counts.

	Not flagged	Flagged
Not fraud	95,004	4,894
Fraud	4	98

True positive = 98 (fraud and flagged). False positive = 4,894 (not fraud and flagged). True negative = 95,004 (not fraud and not flagged). False negative = 4 (fraud and not flagged). The false positives dominate the flagged column, which is the same fact Part 1(a) produced from Bayes' theorem.